

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Adsorption

- Which of the following characteristics is not correct for physical adsorption:-
 (A) Adsorption is spontaneous at suitable conditions
 (B) It is not specific in nature
 (C) It is reversible in nature
 (D) Degree of adsorption increases with temperature
CSF080
- Adsorption is multilayer in case of
 (A) Physical adsorption
 (B) Chemical adsorption
 (C) Both (A) and (B)
 (D) None of these
CSF081
- Which one of the following is applicable to chemisorption-
 (A) It is Irreversible
 (B) Heat of Adsorption is Negative
 (C) It forms unimolecular layer
 (D) All
CSF082
- Which one of the following is not applicable to chemisorption:-
 (A) Heat of chemisorption is negative
 (B) It is reversible
 (C) It forms mono molecular layers
 (D) None
CSF083
- Adsorption of gases on solid surface is generally exothermic because :-
 (A) Free energy increases
 (B) Entropy decreases
 (C) Entropy increases
 (D) Interaction developed between gas and solid particles
CSF084
- Which of the following is true in respect of adsorption ?
 (A) $\Delta H > 0$ $\Delta S < 0$ $\Delta G < 0$
 (B) $\Delta H < 0$ $\Delta S < 0$ $\Delta G < 0$
 (C) $\Delta H > 0$ $\Delta S > 0$ $\Delta G > 0$
 (D) $\Delta H > 0$ $\Delta S > 0$ $\Delta G < 0$
CSF085
- According to Freundlich on increasing the pressure, change in extent of adsorption with respect to pressure (at low pressure range) -
 (A) Constant
 (B) Increase linearly
 (C) Increase but not linearly
 (D) Decreases
CSF087
- Graph between $\log\left(\frac{x}{m}\right)$ and $\log P$ is a straight line inclined at an angle of 45° . When pressure of 0.5 atm and $\log k = 0.699$, amount of solute adsorbed for gram of adsorbent will be :
 (A) 1 gm / g adsorbent
 (B) 1.5 gm / g adsorbent
 (C) 2.5 gm / g adsorbent
 (D) 0.25 gm / g adsorbent
CSF088
- Freundlich adsorption isotherm gives a straight line on plotting:-
 (A) $\frac{x}{m}$ vs P
 (B) $\log \frac{x}{m}$ vs P
 (C) $\log \frac{x}{m}$ vs $\log P$
 (D) $\frac{x}{m}$ vs $\frac{1}{P}$
CSF089

10. When $H_{2(gas)}$ gets adsorbed on transition metals, metal hydride formation takes place. Which of the options will not be correct on the basis of above observation?
 (A) Greater the surface area of transition metals greater will be the extent of adsorption.
 (B) The enthalpy change involved will be in the range of 20-40 kJ/mole.
 (C) The extent of adsorption first increases with increase in temperature then after decrease.
 (D) The adsorption will be irreversible in nature.

CSF096

11. 112 cm^3 hydrogen gas is adsorbed at the surface of 5gm Palladium at 273 K and 2 atm. If the effective surface area of each hydrogen molecule is 0.4 nm^2 , then the specific surface area of Palladium is :
 (A) $2.4 \times 10^6 \text{ m}^2/\text{gm}$ (B) $4.8 \times 10^6 \text{ cm}^2/\text{gm}$
 (C) $1.2 \times 10^4 \text{ cm}^2/\text{gm}$ (D) $1.2 \times 10^4 \text{ m}^2/\text{gm}$

CSF097

12. In vessel O_2 gas undergoes physisorption on adsorbent surface causing its partial pressure to decrease from 6 atm to 'P' atm in a 8.21 L vessel at 500K. If total number of sites is 2×10^{23} on adsorbent surface and every site accommodates effectively three O_2 molecules find P(atm).
 $[N_A = 6 \times 10^{23}]$
 (A) 5 (B) 1 (C) 2 (D) 3

CSF098

Catalysis

13. The process which is catalysed by one of the products is called
 (A) acid-base catalysis (B) autocatalysis
 (C) negative catalysis (D) homogeneous catalysis

CSF090

14. Which reaction show the use of heterogenous catalyst
 (A) $2SO_2(g) + O_2(g) \xrightarrow{NO(g)} 2SO_3(g)$
 (B) $2SO_2(g) + O_2(g) \xrightarrow{Pt(s)} 2SO_3(g)$
 (C) $CH_3COOCH_3(l) + H_2O(l) \xrightarrow{HCl(l)} CH_3COOH(aq.) + CH_3OH(aq.)$
 (D) All of the above

CSF091

15. Promoters and Poison are the substance use in chemical reaction which
 (A) Enhance and decrease the activity of catalyst respectively
 (B) Decrease and enhance the activity of catalyst respectively
 (C) Do not have any effect on catalyst
 (D) Can be used in place of catalyst whenever required

CSF092

16. Catalyst used in hydrogen of Vegetable oil into Vegetable ghee is :
 (A) Ni (B) Fe (C) Pd (D) Pt

CSF093

17. Which of the following catalysts is used in contact process?
 (A) V_2O_5 (B) Fe (C) Co (D) Ni

CSF094

Colloids and their Classifications

18. Smoke is a colloidal dispersion of :-
 (A) gas in solid (B) solid in gas (C) gas in gas (D) liquid in gas

CSF100

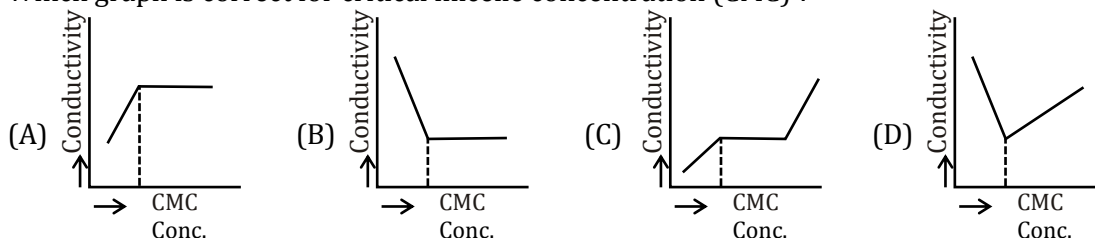
19. Multimolecular colloids are present in :-
 (A) soap solution (B) sol of proteins (C) sol of gold (D) all of these

CSF101

20. The volume of a colloidal particle V_c , volume of a solute particle in a true solution V_t , the volume of suspension particle is V_s can be arranged
 (A) $V_c = V_t = V_s$ (B) $V_s < V_c < V_t$ (C) $V_s > V_c > V_t$ (D) $V_c > V_s > V_t$

CSF102

21. Which graph is correct for critical micelle concentration (CMC) ?



CSF103

22. Which of the following is an example of associated colloid ?
 (A) Protein + water (B) Soap + water (C) Rubber + benzene (D) $\text{As}_2\text{S}_3 + \text{Fe}(\text{OH})_3$

CSF104

23. Milk is an example of
 (A) emulsion (B) suspension (C) foam (D) sol

CSF105

24. Which one of following statements is not correct in respect of lyophilic sols ?
 (A) There is a considerable interaction between the dispersed phase and dispersion medium
 (B) These are quite stable and are not easily coagulated
 (C) They need stabilizing agent
 (D) The particle are hydrated

CSF106

25. Point out the false statement :-
 (A) Colloidal sols are homogeneous
 (B) Colloids carry + ve or - ve charge
 (C) Colloids show tyndall effect
 (D) The size range of colloidal particles is 1-1000 nm

CSF109

Properties of Colloids

26. Point out the false statement :-
 (A) Brownian movement and Tyndall effect is shown by colloidal systems
 (B) Gold number is a measure of the protective power of a lyophilic colloid
 (C) The colloidal mixture of a liquid in liquid is called gel
 (D) Hardy-Schultz rule is related with coagulation

CSF110

27. Select the incorrect statement :-
 (A) Brownian movement depends on the size of the particles
 (B) Lesser the viscosity faster in the motion
 (C) Tyndall effect is not depending upon the diameter of the dispersed particles
 (D) None of these

CSF112

28. Isoelectric point is defined as :-
 (A) The pH at which mobility of colloidal particle is zero
 (B) The pH at which mobility is maximum
 (C) The pH at which mobility is minimum
 (D) None of these

CSF113

29. Tyndall effect would be observed in a
(A) solution (B) solvent (C) precipitate (D) colloidal sol.

CSF114

Properties of Colloids

30. Which of the following ions will be most effective in coagulating the As_2S_3 sol :

(A) Fe^{3+} (B) Ba^{2+} (C) Cl^- (D) PO_4^{3-}

CSF111

31. The ability of an ion to bring about coagulation of a given colloid depends upon
(A) its charge (B) the sign of the charge alone
(C) the magnitude of the charge (D) both magnitude and sign of charge

CSF115

32. An arsenious sulphide sol. carries a negative charge. The maximum precipitating power of this sol. is possessed by.

(A) K_2SO_4 (B) CaCl_2 (C) Na_2PO_4 (D) AlCl_3

CSF116

33. The migration of colloidal particles under the influence of an electrical field is known as
(A) electro osmosis (B) electrophoresis
(C) electrodialysis (D) None

CSF117

34. The arsenious sulphide sol has negative charge. The maximum coagulating power for precipitating it is of :-

(A) 0.1 N $\text{Zn}(\text{NO}_3)_2$ (B) 0.1 N Na_3PO_4 (C) 0.1 N ZnSO_4 (D) 0.1 N AlCl_3

CSF119

Emulsion; Colloids around us and Applications of Colloids

35. An emulsion is a colloidal system of
(A) two solids (B) two liquids
(C) one gas and one solid (D) one gas and one liquid

CSF107

36. At the critical micelle concentration (CMC) the surfactant molecules
(A) decompose (B) dissociate
(C) associate (D) become completely soluble

CSF108

37. When a lyophobic colloidal solution is observed, we can see
(A) light scattered by colloidal particle (B) size of the colloidal particle
(C) shape of the colloidal particle (D) relative size of the colloidal particle

CSF118

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Adsorption

1. These are certain properties related to physical adsorption?
Which of the following properties are for physical adsorption?
- (A) Reversible.
(B) formation of unimolecular layer.
(C) occurs at low temperature and decreases with increasing temperature.
(D) low heat of adsorption.

CSF194

Properties of Colloids

2. The nature of charge on colloidal particles can be experimentally determined by which of the following process?
- (A) Brownian movement (B) Electrophoresis
(C) Electro-osmosis (D) Ultra microscope
3. Which of the following is/are correct statements
- (A) Hardy Schulz rule is related to coagulation
(B) Brownian movement and Tyndall effect are shown by colloids
(C) When liquid is dispersed in liquid, it is called gel.
(D) Gold number is a measure of protective power of lyophilic colloid.
4. Select the Correct Statement(s) -
- (A) Lyophobic sols are irreversible sols.
(B) Micelles formation takes place only above Kraft temperature.
(C) PO_4^{3-} ions have more coagulation value than SO_4^{2-} ions for coagulation of positive sols.
(D) The values of the colligative properties observed experimentally are very small for colloids compare to the true solutions
5. Which of the following are colloids ?
- (A) Milk (B) Ice cream
(C) Urea solution (D) Blood

CSF195

CSF196

CSF197

CSF199

COMPREHENSION TYPE QUESTIONS

Adsorption

Paragraph # 6 to 7

Many lyophilic sols and few lyophobic sols when coagulated under some special conditions changes into semi rigid mass, enclosing whole amount of liquid within itself, it is called gel and the process is called gelation. Gelatin Agar-agar, gum-Arabic can be converted into gels by cooling them under moderate concentration conditions. Hydrophobic sols like silicic acid. $\text{Al}(\text{OH})_3$ are prepared by double decomposition and exchange of solvent method. Types of Gel :

(i) Elastic gel : Those gel which have elastic properties.

Eg : Gelatin, Strach, Agar-Agar etc.

(ii) Non- elastic gel : Those gel which are rigid. Eg : Silica gel. Properties of Gel :

1. Syneresis/weeping of gel : The spontaneous liberation of liquid from a gel is called syneresis or weeping of gels. It is reverse of swelling.

Eg : Gelatin, Agar-Agar show syneresis at low concentration while silicic acid shows it at high concentration.

2. Imbibition or swelling of gel : When gel is kept in a suitable liquid (water) it absorb large volume of liquid. The phenomenon is called imbibition or swelling of gel.

3. Thixotropic : Some gels when shaken to form a sol, on keeping changes into gel are termed as thixotropic gel and phenomenon is called thixotropy.

Eg : Gelatin and silica liquify on shaking changing into corresponding sol and the sol on keeping changes back into gel.

6. Which of the following cannot be used to adsorb water?

(A) Silica gel

(B) Calcium acetate

(C) Hair gel

(D) Cheese

CSF200

7. The process of imbibing water when elastic gel are placed in water is called :

(A) imbibition

(B) syneresis

(C) coagulation

(D) thixotropy

CSF201

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Adsorption

- In a vessel O_2 gas molecules were adsorbed on solid surface causing its partial pressure to decrease from 1 atm to 0.5 atm.

Given : $N_A = 6 \times 10^{23}$

Volume of vessel = 2.24 L

Temperature = 273 K

Total area of solid surface = 10^2 cm^2

Number of active sites per unit area = 10^{24} m^{-2}

Find number of O_2 molecules adsorbed per active site.

CSF120

- A 0.03 M of an acid solution in benzene is dropped on water surface, the benzene evaporates and the acid forms a monomolecular film of solid type. What volume (in ml) of the above solution would be required to cover a 54000 cm^2 surface area of water with monomolecular layer of acid. Area covered by single acid molecule is 0.3 nm^2 ($N_A = 6 \times 10^{23}$)

CSF121

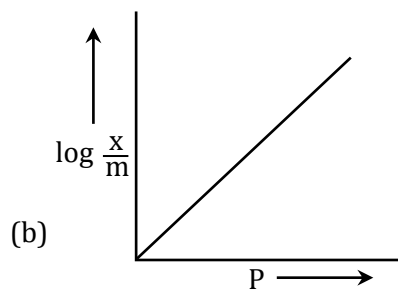
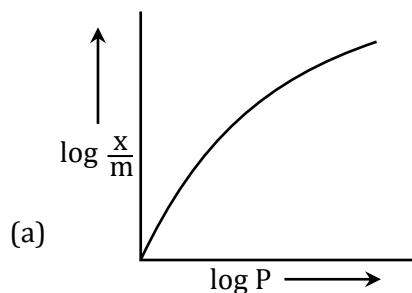
- A container contains 1 litre, 2M solution of benzene in ether. A piece of 3 kg charcoal is dipped in the solution. Molecules of benzene get adsorbed on the surface of charcoal and form monolayer. The molarity of resulting solution decreases to 1M. If surface area available for adsorption on charcoal $12\sqrt{3} \text{ cm}^2/\text{gm}$. Then find distance (in pm) between two adjacent carbon atoms in a benzene molecule.

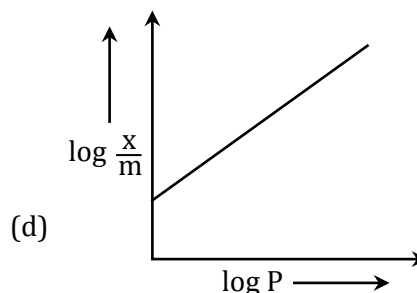
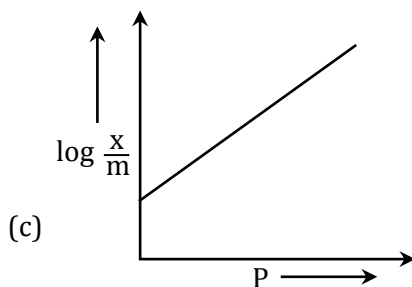
(Assume: Shape of benzene molecule perfectly hexagonal.)

[Use: $N_A = 6 \times 10^{23}$]

CSF123

- Among the following the number of curves not in accordance with Freundlich adsorption isotherm is ____.





CSF202

5. The desorption of gas molecules from the adsorbent surface obeys Arrhenius equation. The average time upto which a N_2 molecule may remain adsorbed at Pt-surface at 400K is
[Given: Pre-exponential factor, $A = 1.25 \times 10^8 s^{-1}$;
Activation energy of desorption = 16Kcal, $e^{20} = 5 \times 10^8$]

CSF125

6. If unit mass of a solid, taken as cube of volume 8 cm^3 , is powdered into identical 10^{12} cubes, the specific surface area of the solid increased by volume of 10^x . then x is :-

CSF099

Properties of Colloids

7. The diameter of a colloidal particle is $5000 \text{ \AA}$. If the density of substance formed, dispersed phase, is 4 gm/cc , find the value of surface area per unit mass of colloidal particle in (m^2/gm).

CSF122

8. A colloid prepared by the addition of KI to AgNO_3 solution is purified using dialysis. Find the minimum mass in grams of an electrolyte 'AB' (GMM = 60) required to completely coagulate 1 l of the aforementioned colloid

Given :

Active ion causing flocculation	Flocculation value(mmole/l)
A^+	50
B^-	100

CSF126

9. If 0.2 gram of an electrolyte AB ($M_w = 200$) is needed for complete coagulation of 200 ml As_2S_3 sol in 2 hr, then what is coagulation value of the electrolyte AB

CSF203

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Adsorption

10. 100 mL of 0.3 M acetic acid shaken with 0.8 g wood charcoal. The final concentration of acetic acid in the solution after adsorption is 0.125 M . The mass of "acetic acid adsorbed per gram of charcoal is x. Then find y (nearest integer), $y = x \times 10$.

CSF204

11. CH_4 is adsorbed on 1 g charcoal at 0°C following the Freundlich adsorption isotherm. 10.0 mL of CH_4 is adsorbed at 100 mm of Hg, whereas 15.0 mL is adsorbed at 200 mm of Hg. The volume of CH_4 adsorbed at 300 mm of Hg is 10^x mL. The value of x is _____ $\times 10^{-3}$.
[Use $\log_{10} 2 = 0.30$, $\log_{10} 3 = 0.48$] (Nearest integer)

CSF205

12. CO_2 gas adsorbs on charcoal following Freundlich adsorption isotherm. For a given amount of charcoal, the mass of CO_2 adsorbed becomes 64 times when the pressure of CO_2 is doubled. The value of n in the Freundlich isotherm equation is _____ $\times 10^{-2}$.
(Round off to the Nearest Integer)

CSF206

13. 3.6 g of oxygen is adsorbed on 1.2 g of metal powder. Volume of oxygen (in litres) adsorbed per gram of adsorbent at STP is Z, then $10 \times Z$ value is

CSF207

14. 112 cm^3 hydrogen gas is adsorbed uniformly at the surface of 5 gm Palladium at 273°C and 2 atm. If the effective surface area of each hydrogen molecule is 0.4 nm^2 , the specific surface area of Palladium is $x \times 10^6 \text{ cm}^2/\text{gm}$. Then find x.

CSF124

15. At 70K, the adsorption of N_2 gas at iron surface obeys Freundlich adsorption isotherm. The experimental data collected is

P(bar)	4	25	64
$\frac{x}{m}$	0.2	0.5	0.8

Where $\frac{x}{m}$ is the mass (in gm) of N_2 gas adsorbed per gm of iron at P bar pressure? The moles of N_2 gas adsorbed per gm of iron at 36 bar and 70 K, is

CSF127

16. If unit mass of a solid, taken as cube of volume 8 cm^3 , is powdered into identical 10^{12} cubes, the specific surface area of the solid increased by 10^x then find value of x.

CSF129

Properties of Colloids

17. $1.9 \times 10^{-4} \text{ gm}$ of the metal having density 19 gm/ml is dispersed in one litre of water to give a sol having spherical metal particles of radius 10 nm . The approximate number of metal sol particles per cm^3 of the sol is given by $x \times 10^9$ then find value of x.

CSF128

18. 20 mL of 0.5 M NaCl is required to coagulate 200 mL of As_2S_3 solution in 2 hours. The coagulating value of NaCl is _____

CSF208

19. For the coagulation of 80 ml of ferric hydroxide sol, 20 ml of 0.25M KCl is required. The coagulation value of KCl will be :-

CSF209

20. The flocculation value of HCl for arsenic sulphide sol. is 30 mmole L^{-1} . If H_2SO_4 is used for the flocculation of arsenic sulphide, the amount, in grams, of H_2SO_4 in 250 ml required for the above purpose is _____.
(molecular mass of $\text{H}_2\text{SO}_4 = 98 \text{ g/mol}$)

CSF210

EXERCISE - JEE (Advanced) PYQ

1. To form a complete monolayer of acetic acid on 1g of charcoal, 100 mL of 0.5 M acetic acid was used. Some of the acetic acid remained unadsorbed. To neutralize the unadsorbed acetic acid, 40 mL of 1 M NaOH solution was required. If each molecule of acetic acid occupies $P \times 10^{-23} \text{ m}^2$ surface area on charcoal, the value of P is ____.

[Use given data : Surface area of charcoal = $1.5 \times 10^2 \text{ m}^2 \text{ g}^{-1}$; Avogadro's number (N_A) = $6.0 \times 10^{23} \text{ mol}^{-1}$]

[JEE (Advanced) 2024]

CSF211

2. The correct statement(s) about surface properties is (are) [JEE (Advanced) 2017]

- (A) Cloud is an emulsion type of colloid in which liquid is dispersed phase and gas is dispersion medium
- (B) Adsorption is accompanied by decrease in enthalpy and decrease in entropy of the system.
- (C) Brownian motion of colloidal particles does not depend on the size of the particles but depends on viscosity of the solution.
- (D) The critical temperatures of ethane and nitrogen are 563 K and 126 K, respectively. The adsorption of ethane will be more than that of nitrogen on same amount of activated charcoal at a given temperature.

CSF078

3. Methylene blue, from its aqueous solution, is adsorbed on activated charcoal at 25°C. For this process, the correct statement is [JEE (Advanced) 2013]

- (A) The adsorption requires activation at 25°C
- (B) The adsorption is accompanied by a decrease in enthalpy
- (C) The adsorption increases with increase of temperature
- (D) The adsorption is irreversible

CSF076

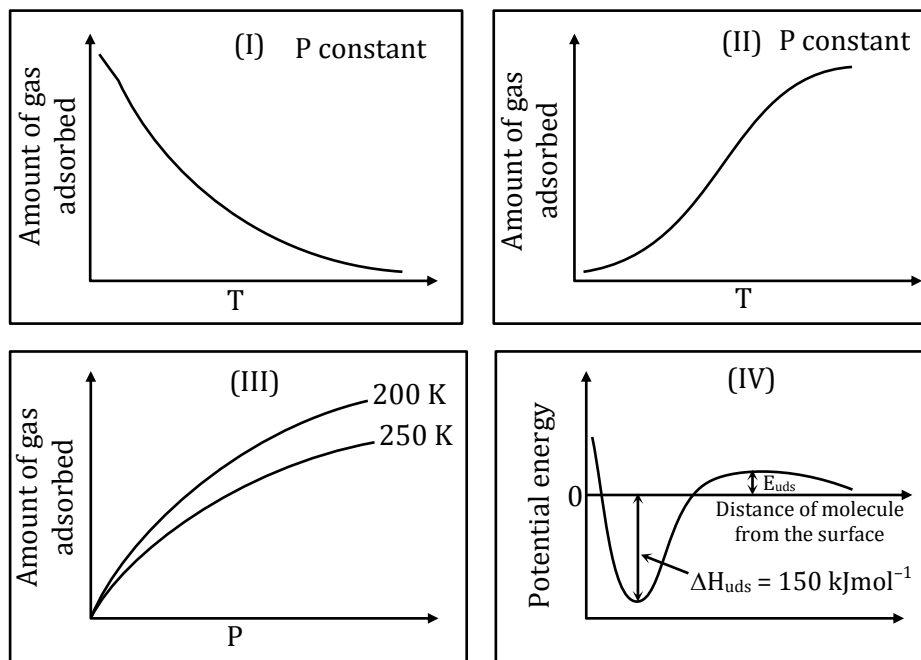
4. Choose the correct reason(s) for the stability of the **lyophobic** colloidal particle.

[JEE (Advanced) 2012]

- (A) Preferential adsorption of ions on their surface from the solution
- (B) Preferential adsorption of solvent on their surface from the solution
- (C) Attraction between different particles having opposite charges on their surface
- (D) Potential difference between the fixed layer and the diffused layer of opposite charges around the colloidal particles

CSF074

5. The given graphs/data I, II, III and IV represent general trends observed for different physisorption and chemisorption processes under mild conditions of temperature and pressure. Which of the following choice(s) about I, II, III and IV is (are) correct? [JEE (Advanced) 2012]



- (A) I is physisorption and II is chemisorption
 (B) I is physisorption and III is chemisorption
 (C) IV is chemisorption and II is chemisorption
 (D) IV is chemisorption and III is chemisorption

CSF075

6. The correct statement(s) pertaining to the adsorption of a gas on a solid surface is (are) –

[JEE (Advanced) 2011]

- (A) Adsorption is always exothermic
 (B) Physisorption may transform into chemisorption at high temperature
 (C) Physisorption increases with increasing temperature but chemisorption decreases with increasing temperature
 (D) Chemisorption is more exothermic than physisorption, however it is very slow due to higher energy of activation

CSF073

7. Consider the following statements related to colloids.

- (I) Lyophobic colloids are **not** formed by simple mixing of dispersed phase and dispersion medium.
 (II) For emulsions, both the dispersed phase and the dispersion medium are liquid.
 (III) Micelles are produced by dissolving a surfactant in any solvent at any temperature.
 (IV) Tyndall effect can be observed from a colloidal solution with dispersed phase having the same refractive index as that of the dispersion medium.

The option with the correct set of statements is

[JEE (Advanced) 2023]

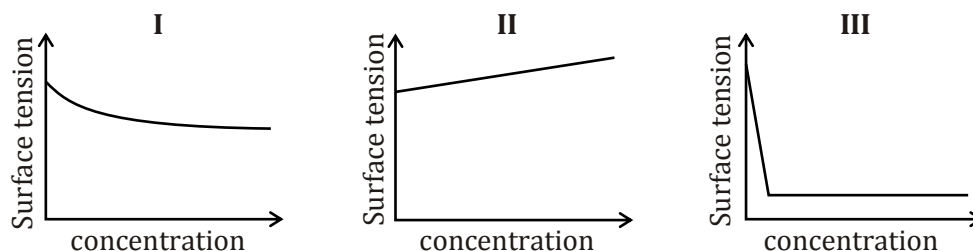
- (A) (I) and (II) (B) (II) and (III) (C) (III) and (IV) (D) (II) and (IV)

CSF212

8. The correct statement(s) related to colloids is(are) [JEE (Advanced) 2021]
 (A) The process of precipitating colloidal sol by an electrolyte is called peptization.
 (B) Colloidal solution freezes at higher temperature than the true solution at the same concentration.
 (C) Surfactants form micelle above critical micelle concentration (CMC). CMC depends on temperature.
 (D) Micelles are macromolecular colloids.

CSF213

9. The qualitative sketches I, II and III given below show the variation of surface tension with molar concentration of three different aqueous solutions of KCl, CH₃OH and CH₃(CH₂)₁₁OSO₃[⊖]Na[⊕] at room temperature. The correct assignment of the sketches is – [JEE (Advanced) 2016]



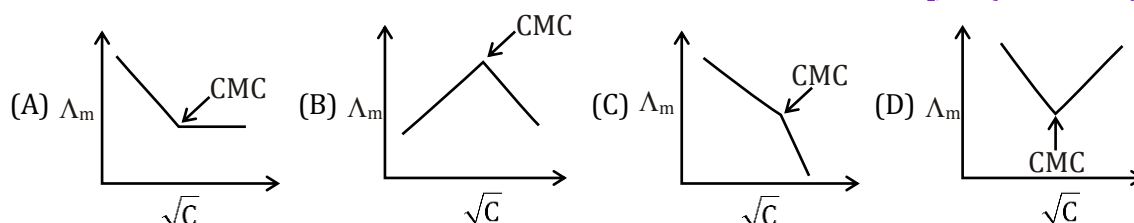
- (A) I : KCl II : CH₃OH III : CH₃(CH₂)₁₁OSO₃[⊖]Na[⊕]
 (B) I : CH₃(CH₂)₁₁OSO₃[⊖]Na[⊕] II : CH₃OH III : KCl
 (C) I : KCl II : CH₃(CH₂)₁₁OSO₃[⊖]Na[⊕] III : CH₃OH
 (D) I : CH₃OH II : KCl III : CH₃(CH₂)₁₁OSO₃[⊖]Na[⊕]

CSF077

10. Among the electrolytes Na₂SO₄, CaCl₂, Al₂(SO₄)₃ and NH₄Cl, the most effective coagulation agent for Sb₂S₃ sol is [JEE (Advanced) 2009]
 (A) Na₂SO₄ (B) CaCl₂ (C) Al₂(SO₄)₃ (D) NH₄Cl

CSF072

11. Molar conductivity (Λ_m) of aqueous solution of sodium stearate, which behaves as a strong electrolyte, is recorded at varying concentration(c) of sodium stearate. Which one of the following plots provides the correct representation of micelle formation in the solution?
 (Critical micelle concentration (CMC) is marked with an arrow in the figures.)



[JEE (Advanced) 2019]

CSF079

12. Among the following, the surfactant that will form micelles in aqueous solution at the lowest molar concentration at ambient conditions is [JEE (Advanced) 2008]
 (A) CH₃(CH₂)₁₅N[⊕](CH₃)₃Br[⊖] (B) CH₃(CH₂)₁₁OSO₃[⊖]Na[⊕]
 (C) CH₃(CH₂)₆COO[⊖]Na[⊕] (D) CH₃(CH₂)₁₁N[⊕](CH₃)₃Br[⊖]

CSF071

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	D	B	D	B	B	C	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	B	B	B	A	A	A	B	C	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	B	A	C	A	C	C	A	D	A
Que.	31	32	33	34	35	36	37			
Ans.	D	D	B	D	B	C	A			

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7			
Ans.	ACD	BC	ABD	ABD	ABD	BCD	A			

EXERCISE (S)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	2	3	4	4	3	6	5	13.00
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1288.00	17.00	21.00	2.40	0.021	4.00	2.35	50.00	62.50	0.36-0.38

EXERCISE - JEE (Advanced) PYQs

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2500	BD	B	AD	AC	ABD	A	BC	D	C
Que.	11	12								
Ans.	C	A								